

01 0:00–0:20 NourishNest

PRODUCT + TECHNICAL DEMONSTRATION

NourishNest

An Agentic AI Household
Nutrition Platform

From pantry awareness to
human-approved meal planning

LOCAL RAG • SPECIALIST AGENTS • HUMAN CONTROL



Welcome to NourishNest, an agentic AI household nutrition platform. It connects everyday food decisions with explainable software: what is available, what could become dinner, and what needs buying. This demonstration follows a fictional family from pantry awareness to human-approved meal planning.

VISUAL: Original reusable diagram. Captions are burned into every timeline frame.
Narration is supplied for your own voice; no audio generated.

02 0:20–0:50 One household, connected workflows

NOURISHNEST / PRODUCT + TECHNICAL DEMO

One household, connected workflows

Everyday value, supported by accountable automation

01 Household profiles

02 Nutrition estimates

03 Recipe collection

04 Pantry awareness

05 Weekly meal planning

06 Grocery preparation

07 Grounded knowledge

08 Specialist agents

09 Human approval + audit

Nine capabilities work together. Household profiles capture preferences and allergies. Nutrition provides estimates, recipes organize ingredients, and the pantry tracks stock and freshness. Planning brings meals into a week, while grocery preparation identifies shortages. Local knowledge retrieval adds cited context. Specialist agents coordinate these services, and human approval controls the final action. The result is a connected workflow with clear explanations and visible boundaries.

FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

02 / 11

VISUAL: Original reusable diagram. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

03 0:50–1:20 How we built it

NOURISHNEST / PRODUCT + TECHNICAL DEMO

How we built it

Layered services keep responsibilities explicit

Streamlit · Household experience



FastAPI · Typed request contracts



Application services · Business rules



Deterministic engines · Calculations



SQLAlchemy + Alembic · Persistence

Agent coordination: supervisor + five specialists

Knowledge: chunks → BM25 → rerank → citations

Structured USDA adapter; optional Ollama.
Both unused in this offline demo.

We built NourishNest in layers. Streamlit presents the experience, FastAPI exposes typed contracts, and application services own business operations. Deterministic engines calculate nutrition, convert units, scale recipes, allocate pantry quantities, and generate grocery needs. SQLAlchemy and Alembic manage persistence and schema evolution. Above those services sit bounded orchestration and local retrieval. This recording uses a rule-based fake provider, not a generative language model. Optional Ollama and live USDA access are unused.

FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

03 / 11

VISUAL: Original reusable diagram. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

04 1:20–1:50 Meet the Greenwood Family

NOURISHNEST / PRODUCT + TECHNICAL DEMO

Meet the Greenwood Family

Fictional profiles; illustrative nutrition

01 Maya + Daniel

02 Vegetarian • Peanut exclusions

03 Saved adult profiles → deterministic estimates

FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

04 / 11

Meet Maya and Daniel in the fictional Greenwood Family. Their saved profiles include vegetarian preferences and peanut allergies. The dashboard connects stock, upcoming meals, and shopping progress. Nutrition estimates come from saved measurements and the versioned calculator, rather than generated prose. Their interest in protein is a planning aspiration, not a promise of nutritional adequacy. All demonstration food values are illustrative, and the application keeps individual dietary suitability subject to human review.

VISUAL: dashboard, household, nutrition. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

05 1:50–2:25 Start with what is already home

NOURISHNEST / PRODUCT + TECHNICAL DEMO

Start with what is already home

Recipes, dated lots and freshness signals

01 7 original demonstration recipes

02 10 foods • 11 lots • 3 locations

03 2 expiring soon • 1 low-stock food

04 FEFO: first-expiring stock is considered first

FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

05 / 11

The recipe collection includes a curry, rice dishes, soup, salad, a tofu stir-fry, and an oat breakfast. Images provide varied serving inspiration, while saved ingredients drive calculations. The pantry has ten foods across refrigerator, freezer, and dry storage. Spinach has two separately dated lots, so quantity alone does not tell the whole story. Freshness signals highlight items worth checking soon, and a tofu threshold flags low stock. First-expiring-first-out logic uses dated inventory when calculating availability; browsing never silently consumes food.

VISUAL: recipes, pantry. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

06 2:25–2:55 Turn inspiration into a plan

NOURISHNEST / PRODUCT + TECHNICAL DEMO

Turn inspiration into a plan

Portions → requirements → pantry coverage → shortages

01 Two dinner portions

02 Explainable pantry coverage

03 Session-local weekly plan

04 Planning does not reserve or consume stock

In Meal Planner, recommendations explain pantry coverage and missing ingredients. We place two portions of lentil and tomato soup into Monday dinner. Changing servings changes the required ingredient quantities and calculated nutrition. The weekly board is session-local, so it is distinct from durable grocery records. Preparing grocery needs combines planned portions and subtracts usable pantry availability. These are point-in-time estimates: a plan does not reserve stock, consume ingredients, or mark anything purchased.

FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

06 / 11

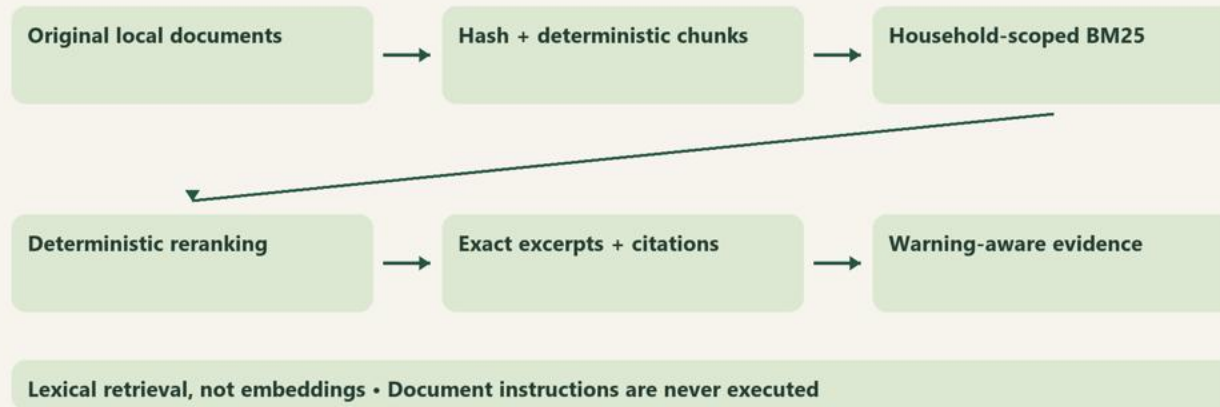
VISUAL: planner. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

07 2:55–3:30 Grounded knowledge, visible evidence

NOURISHNEST / PRODUCT + TECHNICAL DEMO

Grounded knowledge, visible evidence

Deterministic local retrieval — no embeddings



FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

07 / 11

For local knowledge, we ask how cooked rice should be stored safely. An original fictional household note is ingested deterministically, hashed, and split into chunks. BM25 retrieves lexical candidates, reranking orders the evidence, and citations preserve the exact excerpt and source heading. The note deliberately supplies no authoritative storage limits. A separate injection exercise produces an untrusted-instructions warning. Retrieval respects household scope and never executes document instructions. This foundation uses lexical retrieval, not embeddings, and neither RAG nor USDA structured food data replaces deterministic nutrition math.

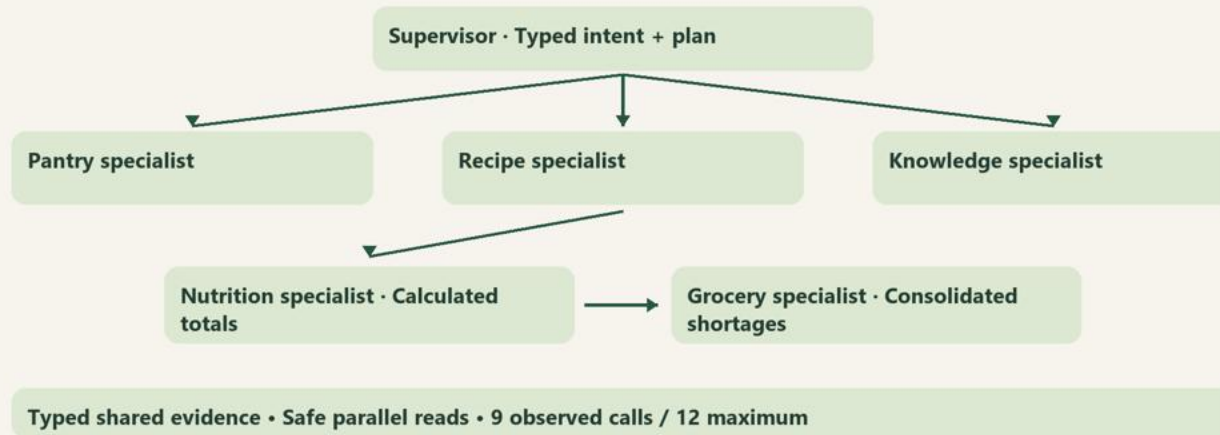
VISUAL: rag-evidence. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

08 3:30–4:25 A bounded team of specialists

NOURISHNEST / PRODUCT + TECHNICAL DEMO

A bounded team of specialists

Six agents • Explicit permissions • At most 12 tool calls



FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

08 / 11

Now the central agentic workflow: create five vegetarian dinners for two adults, prioritize food expiring this week, stay under thirty-five minutes, avoid peanuts, and show grocery shortages. The supervisor translates supported wording into typed intent and an execution plan. Pantry, recipe, and knowledge specialists gather independent evidence with safe read concurrency. Nutrition calculates selected-meal totals, and the grocery specialist consolidates shortages. Six agents operate within explicit tool permissions and a twelve-call ceiling. Typed shared state carries evidence between stages, and the trace records which agent used which tool. Unsupported requests can require clarification or refusal. The output is a reviewable preview, with constraints and warnings visible. It has not created a shopping list, reserved pantry stock, or made a purchase.

VISUAL: assistant-request, assistant-preview. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

09 4:25–5:10 The human controls the write

NOURISHNEST / PRODUCT + TECHNICAL DEMO

The human controls the write

Critic → immutable proposal → approve → execute → audit

1. Critic review

→ 2. Hashed proposal

→ 3. Human approval

→ 4. Execute

Review exact quantities. Approval alone leaves domain data unchanged.

Transaction + version checks. Same retry key returns the same result.

Audit links the decision and controlled write • No purchasing or pantry consumption

FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

09 / 11

The critic checks the preview against constraints and current deterministic calculations, surfacing what remains unverifiable. A proposal freezes the exact list and quantities under a payload hash. The person reviews that payload and explicitly approves it. Approval alone does not create the list: execution is a separate action. Optimistic concurrency rejects stale versions, and transaction boundaries protect against partial writes. Replaying the same execution key returns the same completed result instead of creating another list. The audit trail links the preview, decision, and controlled execution. This local foundation does not authenticate the approving person's identity, and pantry stock and purchase status remain unchanged.

VISUAL: critic, proposal, approved, completed, grocery-detail, execution-evidence. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

10 5:10–5:40 Quality is evidence, not a slogan

NOURISHNEST / PRODUCT + TECHNICAL DEMO

Quality is evidence, not a slogan

Recorded phase results; limited to their evaluated fixtures

01

RAG: 36 evaluation cases

02

Multi-agent: 54 evaluation cases

03

Approval: 64 evaluation cases

04

Last full run: 864 passed / 1 failed

05

Failure corrected; 24 affected tests passed

FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

10 / 11

Quality evidence spans retrieval, orchestration, and controlled execution. Recorded evaluations cover thirty-six RAG cases, fifty-four multi-agent cases, and sixty-four approval cases. The last full suite recorded eight hundred sixty-four passes and one failure; that check was corrected, and twenty-four affected tests then passed. These are fixture results, not a blanket production guarantee. Request identifiers, typed contracts, rollback tests, migrations, and concurrency checks support the foundation. SQLite is demonstrated locally; PostgreSQL remains a deployment target.

VISUAL: Original reusable diagram. Captions are burned into every timeline frame. Narration is supplied for your own voice; no audio generated.

11 5:40–6:00 Grounded. Explainable. Human-approved.

NOURISHNEST / PRODUCT + TECHNICAL DEMO

Grounded. Explainable. Human-approved.

NourishNest

01 **Pantry awareness → Cited evidence → Controlled action**

02 **AI coordination with deterministic accountability**

NourishNest demonstrates how agentic AI can safely coordinate household nutrition workflows while deterministic services retain control of calculations and writes. It brings household context, grounded evidence, and accountable action into one experience. Grounded. Explainable. Human-approved.

FICTIONAL DATA • OFFLINE FAKE PROVIDER • NO PURCHASES

11 / 11

VISUAL: Original reusable diagram. Captions are burned into every timeline frame.
Narration is supplied for your own voice; no audio generated.